

Variable Definitions and Data Sources

	Variable	Description	Freq.	Formula / Purpose	Source
risk_off	Risk-off indicator: 1 when VIX is at least 10 points above its 60-day MA	Daily	$VIX \geq MA60(VIX) + 10$	identifies distress episodes as VAR shock variable	Bloomberg, LSEG
spread	Yield spread between 10-year JGB and 10-year US Treasury	Daily	$JGB10Y - US10Y$	safe-haven bond market channel	Bloomberg, LSEG
log_nikkei	Log of Nikkei-225 stock index	Daily		Equity market response to risk-off, transmission to portfolio flows	Bloomberg, LSEG
log_reer	Log of real effective exchange rate, broad index (2010 = 100)	Monthly → daily		Exchange rate competitiveness channel, safe-haven appreciation	BIS
log_rgdp	Log of constant-price real GDP	Quarterly → daily		Real spillover channel: output response to risk-off	IMF
log_wui	Log of World Uncertainty Index for Japan	Quarterly → daily		Economic policy uncertainty transmission	worlduncertaintyindex.com
debtsec_pct	Net portfolio debt securities inflows (% of nominal GDP)	Monthly → daily	$(debt_flows_usd / GDP_usd) \times 100$	portfolio debt channel	BOJ, IMF, Bloomberg, LSEG
equity_pct	Net portfolio equity inflows (% of nominal GDP)	Monthly → daily	$(equity_flows_usd / GDP_usd) \times 100$	portfolio equity channel	BOJ, IMF, Bloomberg, LSEG
other_pct	Net other investment inflows (% of nominal GDP)	Monthly → daily	$(other_flows_usd / GDP_usd) \times 100$	banking-flow channel	BOJ, IMF, Bloomberg, LSEG
direct_pct	Net direct investment inflows (% of nominal GDP)	Monthly → daily	$(direct_flows_usd / GDP_usd) \times 100$	FDI channel	BOJ, IMF, Bloomberg, LSEG